

Week 4: Gemini Fallback and Rescue Workflow

Project: RESQAI — AI-Assisted Humanitarian Disaster Message Classification

Gemini Fallback

The fourth week focused on handling uncertain predictions. When BERTweet produces a confidence score below 0.70, the message is forwarded to Gemini for contextual analysis. Gemini can provide category, urgency, location, and an explanation.

Low-Confidence Example

For the message “Something happened near the bridge,” BERTweet may return `other_relevant_information` with confidence 0.54. Since 0.54 is below 0.70, the message is sent to Gemini for further contextual analysis.

Not-Humanitarian Verification

The fallback is also triggered when BERTweet predicts `not_humanitarian`, even if its confidence is high. For example, “Beautiful weather today!” may be sent to Gemini for verification. This provides an additional check before the final classification is stored.

Rescue Team Workflow

The rescue-management workflow connects reporter submissions, ML classification, administrative assignment, rescue-team action, and resolution. Admins assign incidents to rescue teams, and rescue teams update incident status. The incident record stores the assigned team, status, priority, notes, and status history.

Security

Security measures include JWT authentication, role-based access control, password hashing, input validation, rate limiting, CORS protection, Helmet security headers, and environment variables for sensitive configuration.

Outcome

By the end of Week 4, the ML decision pipeline and rescue-management workflow were integrated.